

Plutonium

94 Pu Much like uranium, plutonium is used as a nuclear fuel—its most common application.

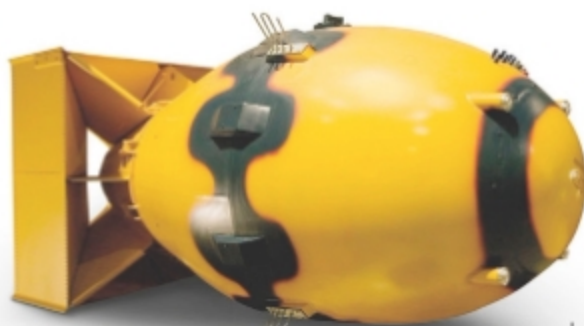
Named after the dwarf planet Pluto, it was the first synthetic element that could be seen with the naked eye. It was discovered in 1940 in Berkeley, California, but its discovery was kept a secret until 1946 because the Americans were concerned about national security and plutonium's use in a nuclear weapon.



Glowing piece of plutonium oxide, the plutonium compound used as a nuclear fuel

Codename: Fat Man

Nuclear weapons were dropped by the US on Japan toward the end of World War II. The plutonium-based "Fat Man" atomic bomb was dropped on the city of Nagasaki in August 1945, just a few days after a uranium-based bomb was dropped on Hiroshima. The devastating nuclear fission reaction that took place in Nagasaki involved plutonium atoms splitting apart to release huge amounts of energy.

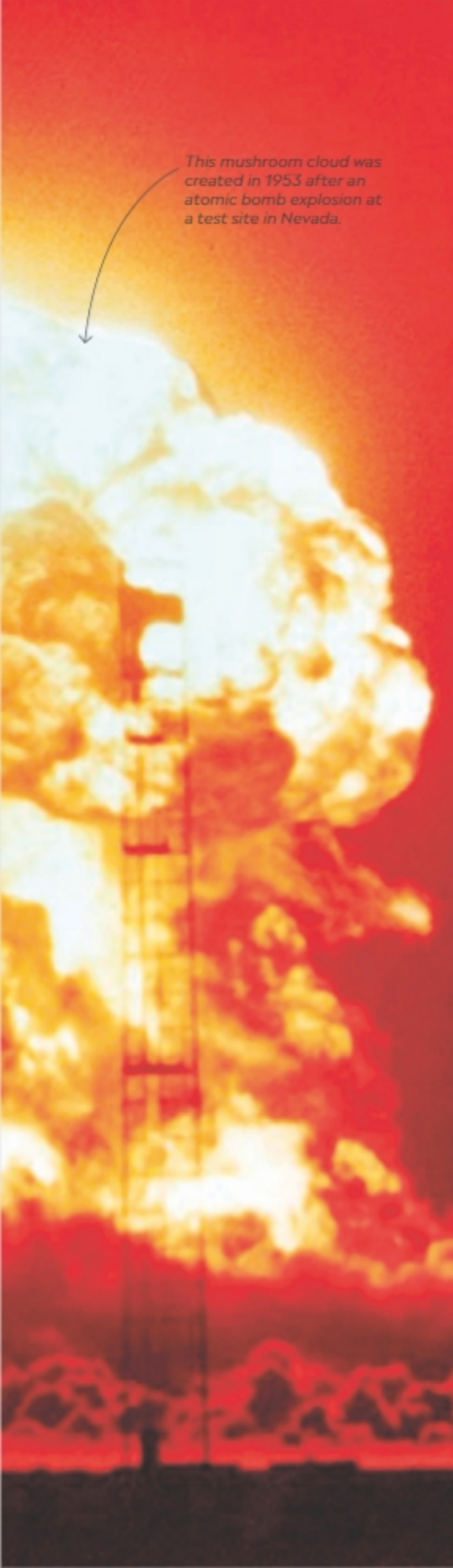


Replica of the "Fat Man" bomb, US Air Force Museum and National Aviation Hall

Artist's representation of the *New Horizons* spacecraft passing Pluto

Far from Earth

Launched in 2006 to study the dwarf planet Pluto, the *New Horizons* spacecraft carried about 22 lb (9.7 kg) of radioactive plutonium—enough fuel to keep its instruments running for 10 years. *New Horizons* made its closest approach to Pluto in July 2015.



This mushroom cloud was created in 1953 after an atomic bomb explosion at a test site in Nevada.